# DNP3

Industrial Control Systems · SCADA Hub

## COURSE TOPICS

1. **Introduction:** Protocol origins, design goals over Modbus, CRC integrity, event-driven reporting, and the typed object model for power utility SCADA.
2. **Protocol Layers:** Application, Transport, and Data Link layers; DNP3's internal three-layer stack and operation on raw serial.
3. **Data Link Layer:** Sync bytes, addressing, control byte, CRC-16 on every 16-byte block, and primary/secondary station roles.
4. **Application Layer:** APDU structure, function codes, object headers, sequence numbers, fragmentation, and multi-fragment exchange.
5. **Data Objects:** Typed object groups and variations — Analog Input Group 30, Binary Input Group 1, quality bytes, and real-deployment object sets.
6. **Function Codes:** Read, Write, Direct Operate, Select-Before-Operate, Freeze, Enable/Disable Unsolicited; SBO two-message sequence and timeout.
7. **Unsolicited Responses:** Class-based event assignment, event buffer sizing, unsolicited confirmation, and master-offline behavior.
8. **Secure Authentication v5:** HMAC challenge/response, pre-shared key management, aggressive mode, supported algorithms, and NERC CIP alignment.
9. **Troubleshooting:** Event buffer overflow, SBO confirm timeout, application layer confirm timeout, and TCP session management on flapping WAN links.
10. **Protocol Lab:** Outstation commissioning, unsolicited reporting setup, SAv5 key management, tshark filtering, and event buffer overflow diagnosis.

## LEARNING OUTCOMES

- Configure an outstation point database, event class assignments, and unsolicited reporting from scratch.
- Trace and verify a Select-Before-Operate sequence in a Wireshark protocol capture.
- Configure Secure Authentication v5 key management on a DNP3 master/outstation link.
- Diagnose event buffer overflow, SBO timeout, and application layer confirm timeout failures.
- Write tshark display filters to isolate DNP3 traffic by function code and event class.

### Certification Relevance

- **GICSP (GIAC)** — DNP3 protocol knowledge and Secure Authentication appear directly in ICS security exam domains; DNP3 is the primary power utility protocol tested.
- **ISA CCST** — SCADA communications and field device protocols tested at CCST Levels II and III.
- **NERC CIP** — SAv5 is a recognized compliance path for CIP-005 and CIP-007 electronic security perimeter requirements.

### Next Steps

1. Take DNP3 alongside the RTAC guide. The RTAC is the most common DNP3 master/outstation in power utility substations.
2. Download Triangle MicroWorks DNP3 Test Harness (free trial) to practice outstation configuration against a software master.
3. IEEE 1815 is the normative DNP3 standard. The DNP3 User Group application notes fill the gaps the standard leaves open.
4. GICSP is the ICS security credential most relevant to power utility and substation SCADA work.